

Rohith Papani

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Machine learning and security engineer focused on privacy-preserving ML, synthetic tabular data generation, differential privacy, membership-inference attacks with hands-on experience benchmarking generative models, building ML/security pipelines, and stress testing generative models to find utility failures.

SKILLS SUMMARY

Languages : C++, python, C, Typescript, Javascript.

ML/Data : PyTorch, scikit-learn, pandas, NumPy, XGBoost, CatBoost, Matplotlib, SQL, MongoDB.

Privacy/Security : Differential Privacy, DP-SGD, Opacus, Membership-Inference Attacks, Synthetic Data Generation, GDB, IDA, Memory Exploitation.

Tools : Linux, Git, Conda, Jupyter, PDB, IDA.

RESEARCH & PROJECTS

Privacy and Utility Benchmarking of Tabular Diffusion Models

CDTD, TabSyn, TabDDPM, PrivBayes, DPWGAN | Python, PyTorch, pandas, NumPy, scikit-learn | [RESULTS](#)

- Built a controlled benchmark evaluating synthetic-data utility from 3 SOTA tabular diffusion models (TabDDPM, CDTD, TabSyn) vs. 1 non-diffusion baseline (PrivBayes) across 6 datasets (mixed/categorical/numerical) on classification-regression utility, feature-level fidelity, pairwise correlation, coverage, and generalization. 10 synthetic sets per config with 95% CI. Extended comparison to DPWGAN on real-data benchmarks.
- Found that diffusion models perform exceptionally well over task metrics than non diffusion models but KDE over PCA maps revealed structural distributional failures (mode collapse/coverage) invisible to downstream classification utility for supervised gaussian datasets.
- Established that off-the-shelf DP-SGD does not transfer to tabular diffusion, measured by downstream Balanced accuracy score collapsing from 0.77–0.82 to ~0.50 (random), 5–6 $\times$ degradation in regression RMSE, and mode collapse in distributional fidelity, by adapting TabDDPM under Opacus across multiple privacy budgets and auditing generated distributions feature-by-feature.
- Stress-tested diffusion models across increasing dimensionality, isolating it as the primary failure axis: as dimensions rose, TabDDPM's downstream accuracy and F1 dropped ~20% (32→128 dimensions), while TabSyn degraded both earlier and harder ~30% from 16→128 dimensions.
- Proposed a novel MIA signal for TabDDPM via Membership Inference Attacks (MIA) on training dynamics, surfacing risk that black-box output audits miss with leakage most pronounced in low-data regimes (1K members vs. 1K non-members), reaching TPR@0.1 FPR of 0.53 (Adult), 0.59 (Shopping), and 0.36 (Bank).

Differentially Private Adaptation of FinDiff for Tabular Data Utility

FinDiff tabular diffusion model | Python, PyTorch, Opacus, pandas, NumPy | [CODE](#)

- Adapted FinDiff into a differentially private tabular generator (Opacus, RDP accountant), sweeping ϵ on UCI Adult; adding DP collapsed minority income-class F1 from 0.32 to 0.00 while overall accuracy barely moved (0.79 → 0.76), as the downstream classifier degenerated to a majority-class predictor exposing accuracy as a misleading proxy under DP.
- Identified that strict differential privacy budgets led to distributional failures, such as race-category polarity reversal and artificial density peaks (continuous features (age, education-years) piled onto their min/max boundary), ultimately degrading both predictive utility and feature fidelity.

Airbnb Price Prediction

Python, scikit-learn, XGBoost, CatBoost | [CODE](#)

- Placed 7th of 80 in a class-wide Kaggle RMSE competition; ensembled Random Forest / XGBoost / CatBoost for a 16.4% RMSE improvement over baseline via distribution-based imputation, log-normalization, correlation-based feature selection, and 5-fold CV tuning with early stopping.

WORK EXPERIENCE

Graduate Student Researcher, University of California, Riverside | 01/2025 - 04/2026

- Owned an Experimental stack used for privacy and utility evaluation of 3 diffusion and 1 non diffusion generative model across real and gaussian datasets.
- Demonstrated that low-data regimes induce memorization in the TABDDPM diffusion model measured via fidelity, coverage, and generalization, isolating the attack surface, then exploiting it to construct a novel membership-inference signal.

Teaching Assistant, University of California, Riverside | 08/2025 - 03/2026

- Taught systems security labs (C, GDB, stack/heap overflow, use-after-free, ROP) to 20 students, authoring lab materials from scratch. | [LAB MATERIALS](#)
- Supported a 50-student C++ course and a 15-student data analysis course covering NumPy/pandas pipelines, cleaning, feature engineering, and visualization.

EDUCATION

Doctoral Research - Computer Science (coursework completed) | 09/2024 – 04/2026 | *University of California, Riverside* | **GPA: 3.60/4.00**

Master of Science - Computer Science | 08/2022 - 05/2024 | *Arizona State University* | **GPA: 4.00/4.00**

Bachelor of Technology - Computer Science | 07/2018 - 06/2022 | *Vellore Institute of Technology (VIT) , Vellore* | **CGPA: 8.79/10.00**